



Flexible Public Transport Modelling for Large Urban Areas

Jeroen P.T. van der Gun, Rob van Nes, Bart van Arem



Introduction

- Public transport makes travel demand models complex
 - PT enables many potential combination of transportation modes
 - Constructing a good choice model is cumbersome
- How to create a model that, contrary to currently existing models, is
 - Flexible w.r.t. how modes are valued and used
 - Consistent in the choice process
- Focus on mode and route choice, static modelling

Presentation contents

- Problem description
- Ideal model
 - Interface mode/route choice
 - Route set generation
 - Choice model
- Case study
 - Introduction
 - Route generation
 - Choice model
- Conclusions and recommendations

Problem description (1)

Aggregation of modes

- Different modes are modelled as one joint mode
 - Differences within the group are neglected

Example: GroeiModel

Bus and tram equivalent?

Real modes	Modes in model
Train	Train
Bus	Bus/tram/metro
Tram	
Metro	

Problem description (2)

Addition of new modes

- Aggregate new mode with an existing mode?
- Add a real new mode?
 - Interaction with existing modes?

Example: IJmeerlijn in GroeiModel

- Possible future rail connection Amsterdam-Almere
- Is it train or metro?
 - Determines valuation of the mode
 - Determines possible combinations with other modes



Problem description (3)

Consistency of the choice process

- Logit models based on utility maximisation
- In terms of expectation, a group of options is more attractive than the options on themselves
 - Due to differences in preferences
 - among travellers
 - between travellers and researcher
 - Look out for mutual dependencies between options within the group (positive correlations of utilities)

Example: GroeiModel

- Ignores diversity in possible bus/tram/metro routes
- Ignores route overlap for train routes

Interface mode/route choice (1)

Classification

- Models can be classified according to two dimensions:
 - How are networks combined in a route?
 - How many modes are contained in a network?
- The traveller has two choices in a model:
 - What networks will I use?
 - What route do I choose within these networks?
- Interface needs to be determined...

Interface mode/route choice (2)

Flexibility and consistency

- Three types of positive correlations between utilities of route alternatives

- Route overlap
- Modal overlap
- Mode similarities

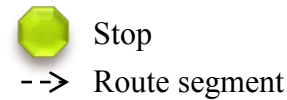
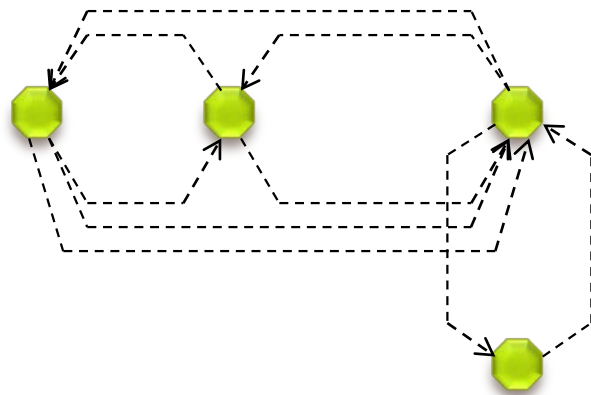
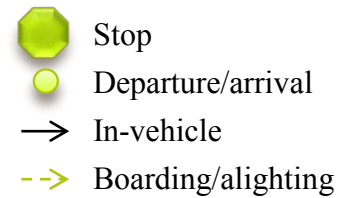
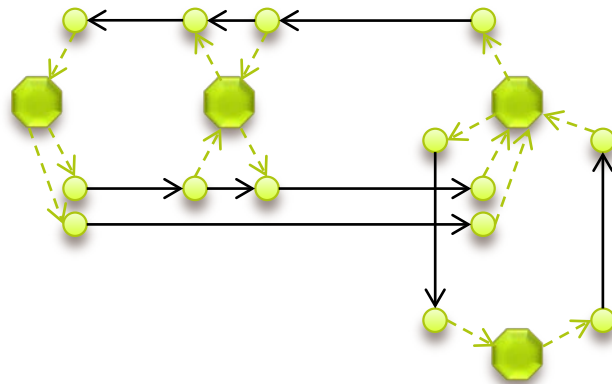
	No modal overlap	Partial modal overlap	Full modal overlap
No route overlap			
Partial route overlap	N/A		
Full route overlap	N/A	N/A	

→ Mode 1
→ Mode 2

- Supernetwork required for flexibility and consistency
 - All modes should form a single network

Route set generation (1)

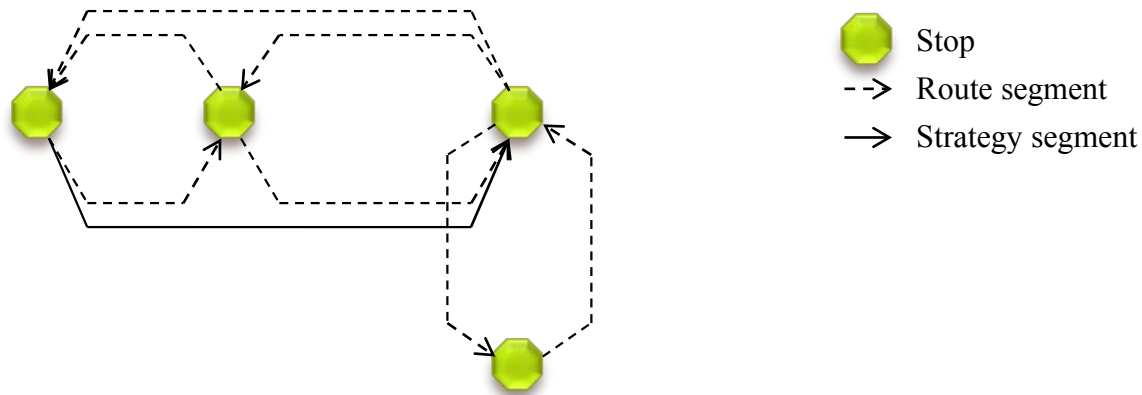
Public transport network structure



Route set generation (2)

Common lines problem

- Common lines can now be merged
 - Simplifies network
 - Reduces the choice set size
 - More realistic from behavioural perspective (if same mode)



Route set generation (3)

Algorithm

- Branch-and-bound algorithm suitable for public transport network
 - Systematically iterate all possible routes within boundaries using the branch-and-bound algorithm
 - Choice set contents explicitly defined by search constraints
 - Tolerance constraint
 - With trade-off between number of legs and travel time
 - Logical constraints
 - Dominance constraint
 - Efficient for this public transport network with merged common lines
 - Number of segments in the network is large
 - Number of segments in a route is small

Route set generation (4)

Completing the supernetwork

- Adding private modes to complete the supernetwork should not increase the number of links per route too much
 - Otherwise, branch-and-bound algorithm will become very inefficient (search tree depth)
- Therefore, find access/direct/egress sub-routes in private mode network using Dijkstra and add these as segments to the supernetwork
 - For uni-modal route choice for private modes, you could use an additional route set generator

Choice model

Network GEV path size logit

Nested logit:

mode similarities

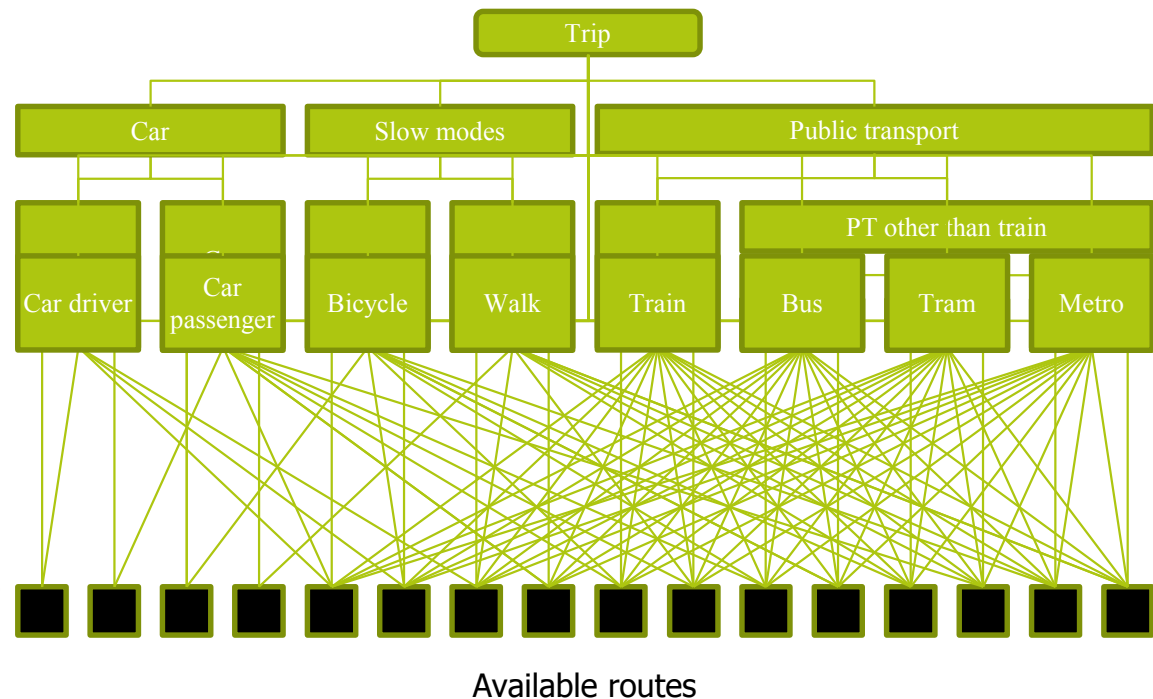
Cross-nested logit:

modal overlap

Path size factors

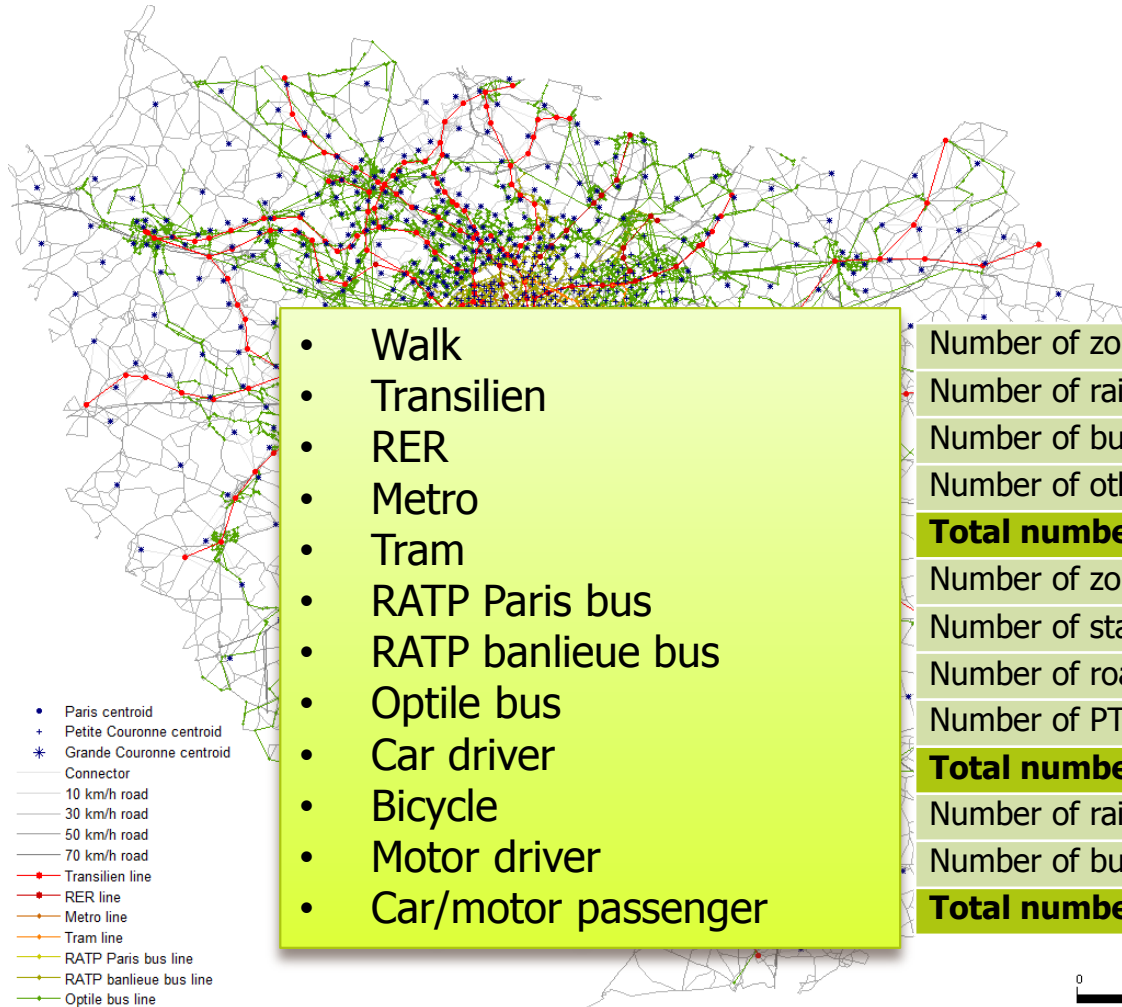
Path size logit:

route overlap



Case study: introduction

Île-de-France



Morning peak

- Walk
- Transilien
- RER
- Metro
- Tram
- RATP Paris bus
- RATP banlieue bus
- Optile bus
- Car driver
- Bicycle
- Motor driver
- Car/motor passenger

Number of zones	1.342
Number of rail stations	936
Number of bus stops	10.978
Number of other road nodes	56.407
Total number of nodes	69.663
Number of zone connectors	21.336
Number of station connectors	10.546
Number of road links	261.518
Number of PT transfer links	15.054
Total number of links	308.454
Number of rail lines	198
Number of bus lines	2.494
Total number of PT lines	2.692



Case study: route generation (2)

Mapping observed routes to the supernetwork

Model is estimated based on household survey Enquête Globale Transport

- Home-work trips in morning peak
- Origins/destinations in Grande Couronne excluded

Route observations need to be mapped to the supernetwork:

What route did you take?

Car to node #2001
Metro to node #2034
Walk to node #10293
Bus to node #11839
Walk to destination

Car to (102,201)
Metro line 12 to (153,241)
Bus to node (211,294)

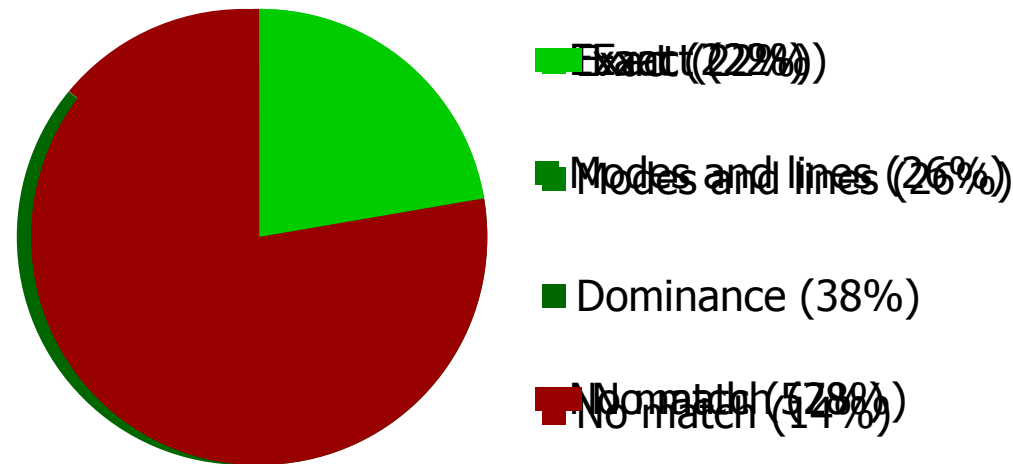
Case study: route generation (3)

Coverage of observed routes

Observed routes can now be compared with generated routes

For model estimation, the coverage of the route generation process is important

- Are routes observed in the survey also generated?
- Otherwise, one cannot choose them in the model



Case study: choice model (1)

Attributes mode and route choice model

- Time in private modes (own vehicle or walk)
- Time in PT modes
- Waiting time (max. 7.5 minutes per boarding)
- PT costs
 - Taking personal discounts into account
- Dummy for using PT without discounts
- Number of legs ("boarding penalties")
 - For each mode separately
 - Separately with/without PT usage
- "Domination size" $\ln(1+n)$
- Vehicle and driving licence ownership taken into account through availability of alternatives

Case study: choice model (2)

Best estimated logit models

		MNL	PSL	NL	NPSL
Log-likelihood		-2857.6	-2853.5	-2835.8	-2837.6
ρ^2		0.437	0.438	0.442	0.441
Observations		2523	2523	2523	2523
Free coefficients		20	21	22	22
β	Private mode time	-7.59 h ⁻¹ (-22.2)	-7.61 h ⁻¹ (-22.3)	-6.90 h ⁻¹ (-12.1)	-6.84 h ⁻¹ (-15.4)
	PT in-vehicle time	-4.11 h ⁻¹ (-10.3)	-4.25 h ⁻¹ (-10.5)	-3.59 h ⁻¹ (-8.5)	-3.60 h ⁻¹ (-9.7)
	PT waiting time	-5.89 h ⁻¹ (-6.0)	-5.80 h ⁻¹ (-5.9)	-6.19 h ⁻¹ (-6.4)	-6.15 h ⁻¹ (-6.9)
	PT costs	-0.52 € ⁻¹ (-4.9)	-0.57 € ⁻¹ (-5.3)	-0.39 € ⁻¹ (-4.1)	-0.41 € ⁻¹ (-4.5)
	PT usage w/o discounts	-2.24 (-9.5)	-2.18 (-9.2)	-2.28 (-7.9)	-2.25 (-8.3)
	Transilien legs	-0.02 (-0.1)	-0.09 (-0.5)	-0.32 (-0.2)	-0.03 (-0.2)
	RER legs	-0.42 (-3.1)	-0.53 (-3.7)	-0.32 (-2.8)	-0.32 (-3.0)
	Metro legs	-0.44 (-6.5)	-0.49 (-7.0)	-0.34 (-6.3)	-0.40 (-7.3)
	Tram legs	-0.17 (-1.0)	-0.21 (-1.2)	-0.15 (-1.1)	-0.21 (-1.5)
	RATP Paris bus legs	-2.30 (-10.0)	-2.44 (-19.2)	-1.74 (-13.5)	-1.75 (-18.5)
	RATP suburbs bus legs	-1.30 (-11.7)	-1.33 (-11.9)	-1.07 (-10.0)	-1.07 (-11.8)
	Optile bus legs	-2.52 (-7.1)	-2.58 (-7.2)	-2.03 (-6.6)	-2.03 (-7.4)
	Access car driver legs	-2.82 (-9.0)	-3.06 (-9.0)	-2.32 (-8.5)	-2.30 (-9.8)
	Direct car driver legs	-2.26 (-15.1)	-2.25 (-15.1)	-2.17 (-9.9)	-2.13 (-11.5)
	Direct motor driver legs	-1.89 (-9.1)	-1.90 (-9.2)	-1.89 (-7.5)	-1.87 (-8.1)
	Acc./egr. bicycle legs	-5.83 (-8.1)	-5.87 (-8.2)	-4.43 (-7.5)	-4.32 (-8.6)
	Direct bicycle legs	-3.83 (-23.8)	-3.80 (-23.6)	-3.67 (-11.2)	-3.61 (-13.7)
	Access passenger legs	-3.90 (-9.7)	-3.92 (-9.7)	-3.09 (-8.7)	-3.02 (-10.5)
	Direct passenger legs	-5.48 (-25.9)	-5.48 (-25.9)	-5.37 (-11.0)	-5.33 (-13.4)
	Domination size	0.16 (5.6)	0.15 (5.1)	0.11 (4.7)	0.10 (4.1)
γ	Path size		-0.34 (-2.8)		-0.28 (-3.1)
θ	PT nest			0.74 (-4.8)	0.68 (-6.9)
	Metro/tram nest			0.57 (-4.1)	

Case study: choice model (3)

Aggregation of modes

In ANTONIN:

	Train		Metro		Bus		
	Transilien	RER	Metro	Tram	RATP Paris bus	RATP banlieue bus	Optile bus
Transilien		+2.1	+2.1	+0.7	+9.9	+7.1	+6.5
RER	-2.1		+0.2	-1.0	+11.3	+7.1	+6.0
Metro	-	-0.2		-1.4	+12.7	+7.7	+5.7
Tram	-0.7	+1.0	+1.4		+9.1	+6.0	+5.9
RATP Paris bus	-9.9	-11.3	-12.7	-9.1		-6.4	+1.0
RATP banlieue bus	-7.1	-7.1	-7.7	-6.0	+6.4		+3.5
Optile bus	-6.5	-6.0	-5.7	-5.9	-1.0	-3.5	

- Significant differences in boarding penalties within buses and within trains
- Aggregation of modes hence is indeed problematic

Case study: choice model (4)

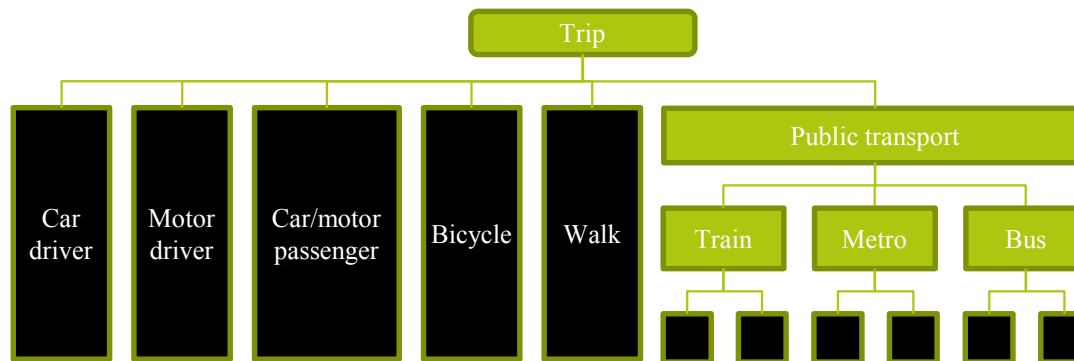
Consistency in choice process: positive correlations

- Route overlap: no positive correlation
 - On the contrary: overlapping routes seem more attractive, possibly due to ad hoc choice behaviour
 - Path size coefficient is significant with the wrong sign
 - However, a negative path size coefficient is no correct model of ad hoc route choice options

Case study: choice model (5)

Consistency in choice process: positive correlations

- Nested path size logit model
 - Necessary simplification due to software limitations



- Modal overlap: significant positive correlation among routes with main mode metro
- Mode similarities: significant positive correlation among routes with PT
- Modal overlap for private modes (different nesting): significant negative correlation among routes with PT where PT part is identical

Conclusions and recommendations

- Theoretical framework developed for flexible PT modelling
- Case study shows feasibility in practice
- Case study supports expected advantages compared to currently existing model structures
- Recommended subjects for further research include:
 - Network loading results
 - Ad hoc route choice behaviour
 - Branch and bound algorithm optimisation
 - Network GEV model usage
 - Timetable information usage
 - Robustness of adding new modes

References

The archived submission did not contain a separately labelled references section.